

Physics-Informed Super-Resolved Ultrasonic Imaging

Local Inverse Modeling and Unrolled Reconstruction

Kadidiatou Diallo, Fred Maurice Ngolè Mboula, Florian Le Bourdais

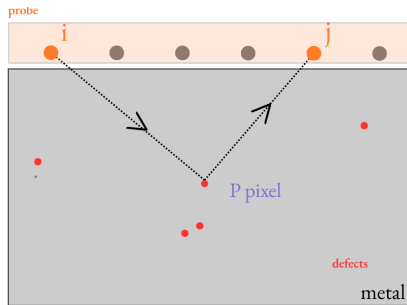
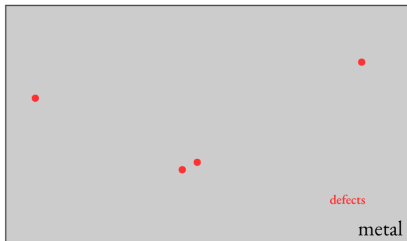
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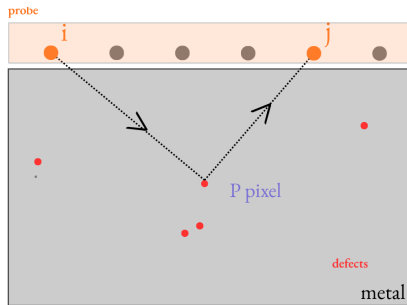
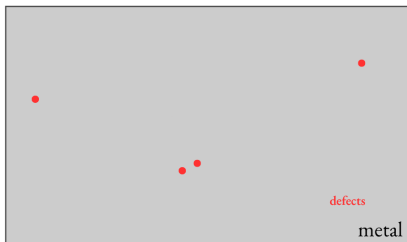
Ultrasonic Imaging for Non-Destructive Testing (NDT)

- **Inspection of critical components:**
 - cracks, porosities, delaminations.
 - aerospace, industrial manufacturing.



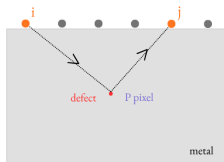
Ultrasonic Imaging for Non-Destructive Testing (NDT)

- **Inspection of critical components:**
 - cracks, porosities, delaminations.
 - aerospace, industrial manufacturing.
- **N -element probe**
 - sequential transmission, reception on all elements.
 - N^2 **A-scans** $s_{ij}(t)$, one per transmitter-receiver pair.
 - image formed by combining the contributions of each pair.

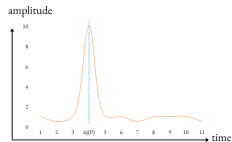


Total Focusing Method (TFM)

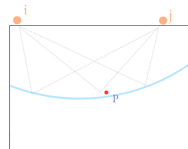
- For a pair (i, j) and a pixel P : round-trip time of flight $t_{ij}(P)$.
- Contribution of the pair: A-scan amplitude at that time, $s_{ij}(t_{ij}(P))$.



Transmitter Receiver Pair (i, j)



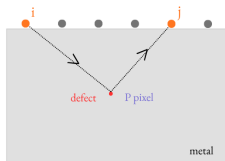
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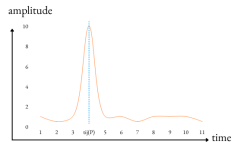
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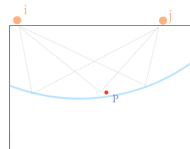
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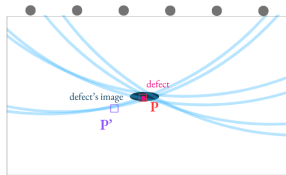
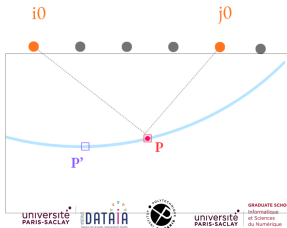


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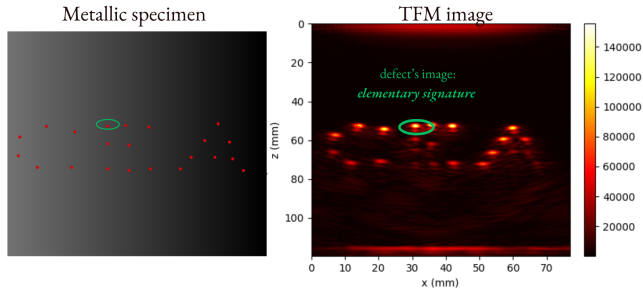
- **Coherent summation:** contributions align at the defect and scatter elsewhere.



$$I(P) = \sum_{i=1}^N \sum_{j=1}^N s_{ij}(t_{ij}(P))$$

TFM Limitations and Positioning

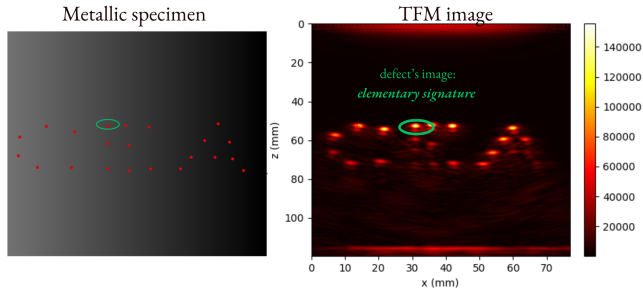
- finite bandwidth: *elementary signature* of non-zero duration.
- spatially spread response.
- closely spaced reflectors cannot be separated.



Rayleigh criterion: $R \propto \frac{l}{\sin \phi}$,
 l wavelength, ϕ angular aperture (set by the depth and the probe aperture)

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- goal: resolution below the Rayleigh limit.
- **physics-informed** approach: interpretable, data-efficient, transferable.

Imagettes: Our Pixel-Wise Local Representation

- At each pixel P , keep the individual contributions $y_{ij}(P) = s_{ij}(t_{ij}(P))$ instead of summing them.

$$I(P) = \sum_{i,j} y_{ij}(P)$$

TFM reduces everything to a scalar.

$$Y(P) = [y_{ij}(P)]_{i,j} \in \mathbb{R}^{N \times N}$$

one $N \times N$ matrix per pixel.

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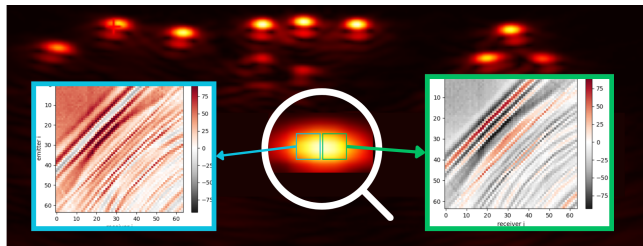
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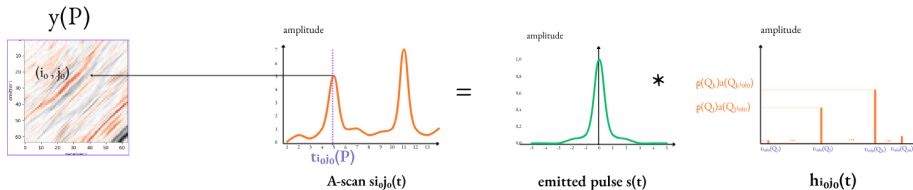
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- Richer*: the information lost by summation is preserved.
- Discriminative*: coherent pattern at the true reflector, incoherent in its neighborhood, for comparable TFM values.



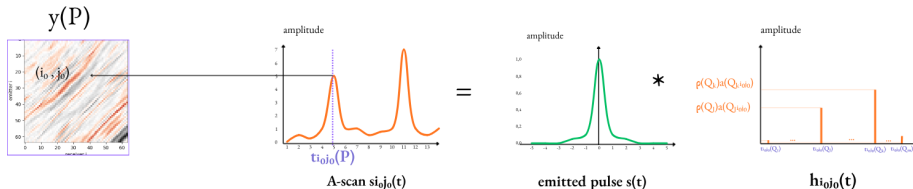
A Local Linear Model

- Each A-scan $s_{ij}(t)$ is the emitted pulse s convolved with the response of the medium.



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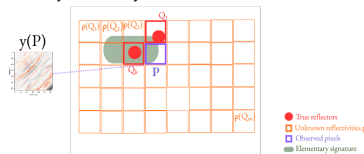
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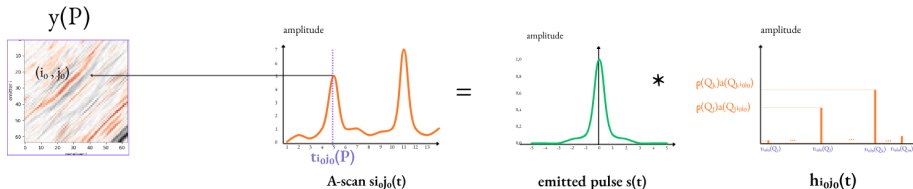
$$y_{ij}(P) = \sum_{Q \in \mathcal{N}(P)} \rho_Q \frac{s(t_{ij}(P) - t_{ij}(Q))}{d_{ij}(Q)^{\frac{1}{\alpha}}}$$

$$\boxed{y(P) = A(P) \rho + \epsilon}, \quad A(P) \in \mathbb{R}^{N^2 \times M}, \quad M = \text{card } \mathcal{N}(P)$$



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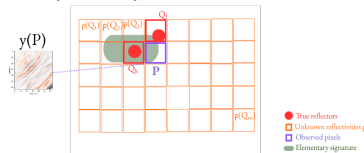
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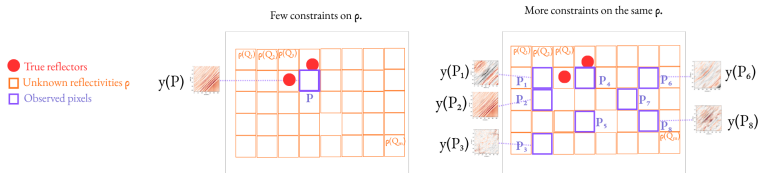
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- Recovering the reflectivity ρ : **ill-conditioned inverse problem**.
 - atoms at neighboring positions are highly correlated.

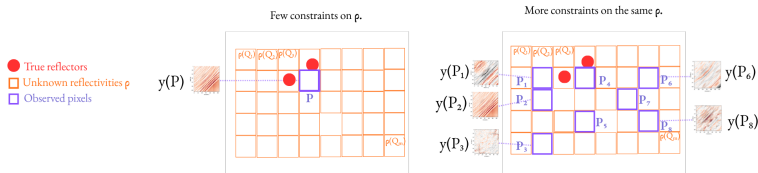
Combining Imagettes Before Inversion

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$$\mathbf{y} = [\mathbf{y}(P_1)^\top, \dots, \mathbf{y}(P_M)^\top]^\top, \quad \mathbf{B} = [A(P_1)^\top, \dots, A(P_M)^\top]^\top, \quad \mathbf{y} = \mathbf{B}\rho + \epsilon.$$

- Each imagette adds new constraints on ρ : $\ker \mathbf{B} = \bigcap_{k=1}^M \ker A(P_k)$, hence a better rank for $\mathbf{B}$ than for a single $A(P)$.

The Local Inverse Problem and Its Spectrum

- *Least-squares* estimation:

$$\hat{\rho} = \arg \min_{\rho} f(\rho), \quad f(\rho) = \frac{1}{2} \|B\rho - \mathbf{y}\|_2^2, \quad \nabla f(\rho) = G\rho - \mathbf{b}, \quad G = B^\top B, \quad \mathbf{b} = B^\top \mathbf{y}.$$

$$\implies \text{normal equations } \boxed{G\rho = \mathbf{b}}, \quad G \in \mathbb{R}^{M \times M}.$$

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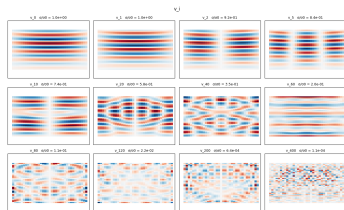
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- Columns of V : **orthonormal basis** of the reflectivity space.



- σ_i^2 : how strongly $\mathbf{v}_i$ is constrained by the measurements.
- **small values**: slow convergence, noise amplification.

Resolution Below the Rayleigh Limit

- ISTA,
$$\hat{\rho} = \arg \min_{\rho \geq 0} \frac{1}{2} \|B\rho - \mathbf{y}\|_2^2 + \lambda \|\rho\|_1.$$
 - *sparsity*: a natural prior in NDT.
- Rayleigh limit: $R = \frac{0.61 \, l}{\sin \phi} = \frac{0.61 \, l}{\sin \arctan\left(\frac{D}{2z}\right)} \approx 0.86 \, l.$
- CIVA data.
 - steel, $N = 64$ -element linear array probe.
 - 5 MHz, $l = 1$ mm.
 - two reflectors at $z = 16$ mm, separated by $0.5 \, l$.

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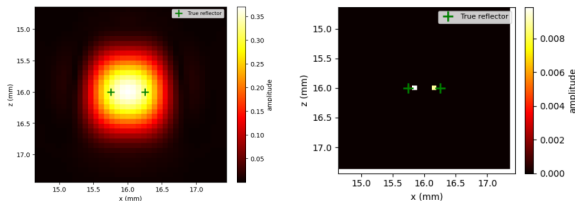
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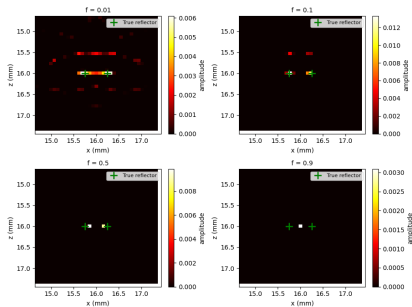
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Separation of $0.5 l$, below the criterion: TFM forms a single lobe, while the inversion separates the two reflectors.

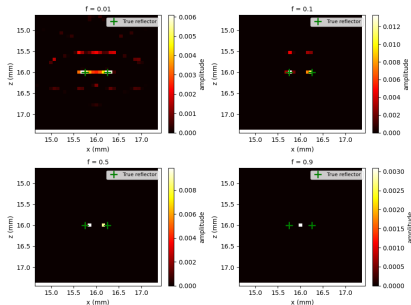
Limitations of Iterative Inversion

- Tuning λ without ground truth.
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- Unexploited directions.

$$\rho^{(k)} = \sum_i c_i^{(k)} \mathbf{v}_i, \quad \rho^{(0)} = \mathbf{0}, \quad c_i^* = \frac{\mathbf{v}_i^\top \mathbf{b}}{\sigma_i^2}$$

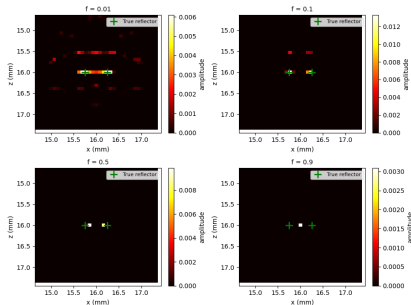
$$c_i^{(k)} = \left(1 - \left(1 - \frac{\sigma_i^2}{\sigma_{\max}^2}\right)^k\right) c_i^*$$

- $\sigma_i/\sigma_{\max} = 10^{-2}$: about 10^4 iterations.
- at the rank threshold, about 10^7 .

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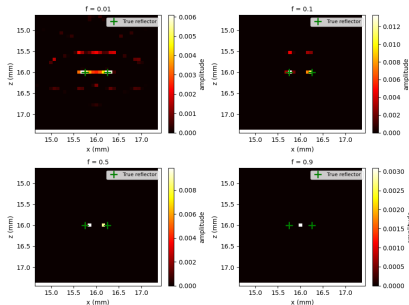
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$$\hat{\rho} = \arg \min_{\rho \geq 0} \frac{1}{2} \|\mathbf{y} - B\rho\|_2^2 + \lambda \|\rho\|_1 + \frac{\gamma}{2} \rho^\top L \rho \quad \text{to tune: } \lambda, \gamma, \varepsilon$$

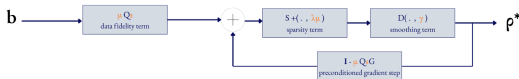
Each mechanism adds a hand-tuned parameter, while λ alone is already a problem.

From an Iteration to a Layer

- **Preconditioned gradient step:** *linear*.

$$\mathbf{z}^{(k)} = \boldsymbol{\rho}^{(k)} - \mu_k \mathbf{Q}_{\varepsilon_k} (\mathbf{G} \boldsymbol{\rho}^{(k)} - \mathbf{b}) = \underbrace{(\mathbf{I} - \mu_k \mathbf{Q}_{\varepsilon_k} \mathbf{G})}_{\text{weights } \mathbf{W}_k} \boldsymbol{\rho}^{(k)} + \underbrace{\mu_k \mathbf{Q}_{\varepsilon_k} \mathbf{b}}_{\text{bias}}$$

- **Thresholding** $\mathcal{S}_{\lambda_k}^+(\mathbf{x}) = \max(\mathbf{x} - \lambda_k, 0)$: ReLU with threshold λ_k , *nonlinear*.
- **Diffusion** $\mathcal{D}_k(\mathbf{x}) = \mathbf{x} - \gamma_k \mathbf{L}(\mathbf{x}) \mathbf{x}$: *nonlinear*.

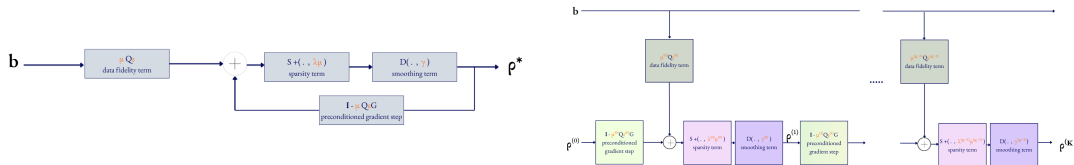


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- **Affine** layer followed by a **nonlinearity**, but W_k is **fixed** by G and the bias = $\mathbf{b}$.
- *Bilevel optimization:*

$$\hat{\rho}_n = \mathcal{A}_K(\mathbf{y}_n; \lambda, \gamma, \mu, \varepsilon) \quad (\hat{\lambda}, \hat{\gamma}, \hat{\mu}, \hat{\varepsilon}) = \arg \min_{\lambda, \gamma, \mu, \varepsilon} \sum_n \mathcal{L}(\hat{\rho}_n, \rho_n)$$

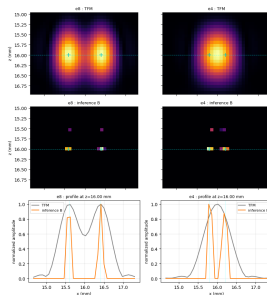
Reconstruction After Training

- Supervised training, 50 layers.
 - 5000 *inverse crime* samples, point-like and extended.
 - a few *CIVA* simulations.



Reconstruction After Training

- Supervised training, 50 layers.
 - 5000 *inverse crime* samples, point-like and extended.
 - a few *CIVA* simulations.
- Inference: two reflectors separated by 0.8 and 0.4 mm, below Rayleigh ($R \approx 0.86$ mm).



TFM: dip at 0.58 of the maximum, then a single lobe.
Network: two peaks at the exact positions, dip down to zero.

Conclusion

- **Local physics-based inversion**, no training: separation below Rayleigh.
- **Unrolled network**: explicit operator per layer, six learned scalars, no manual tuning.
- Interpretable, data-efficient, transferable.
- **Next steps**: directivity, extended defects, off-grid formulation.

Encode the known physics, learn only the rest.

Appendix: Inversion Strategies and Benchmark

- Two regimes: *inverse crime* ($\mathbf{b} = G\rho_{\text{GT}}$) and *CIVA*.

- Inversion strategies:

- Pseudo-inverse**

→ unstable.

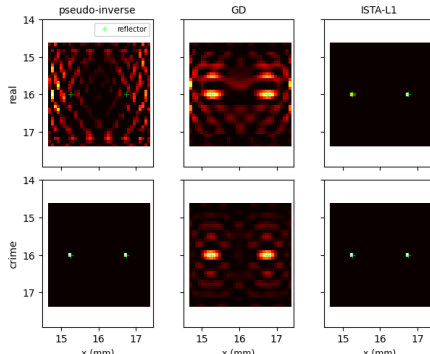
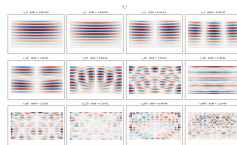
- Projected gradient descent**

→ stabilized by early stopping.

→ fine structures lost.

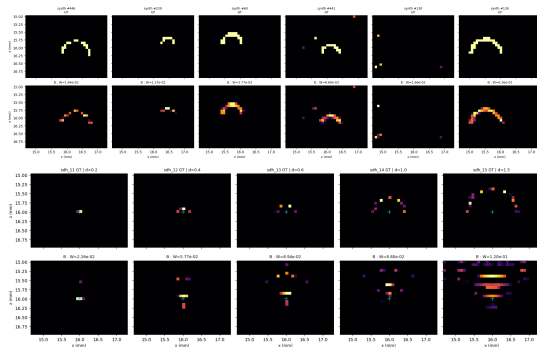
- ISTA** (*Iterative Shrinkage-Thresholding Algorithm*)

→ fine structures can be represented.



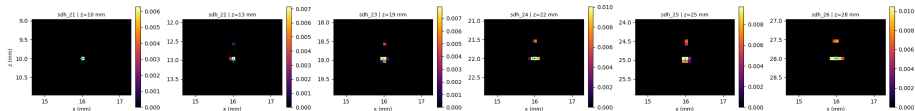
Appendix: Current Limitation: Extended Defects

- *Synthetic*: arcs well reconstructed.
- *CIVA*: degradation as the diameter increases.
 - extent recovered, but with spurious contributions.
- Cause: **non-isotropic reflection**.
$$\Delta \mathbf{b} = \mathbf{B}^\top (\mathbf{B}_{\text{dir}} - \mathbf{B}) \boldsymbol{\rho}$$
 - zero for isotropic reflection.
 - grows with the surface orientation.
- **Ongoing work**: directivity in the forward model and in the training data.



Appendix: Transfer to Unseen Depths

- Network trained at $z = 16$ mm, applied without retraining.
 - $\mathbf{G}$ and $\mathbf{b}$ recomputed for each depth.



- Correct localization from 10 to 28 mm, compact reconstruction from 13 to 22 mm.
- Outlook: a grid of networks trained once, with pre-detection to position the neighborhoods.

Appendix: Computational Aspects

- Sizes ($N = 64$, $M = 1225$): $B \in \mathbb{R}^{MN^2 \times M}$, about $5 \cdot 10^6$ rows, while $G \in \mathbb{R}^{M \times M}$.
- B is never formed, G and $\mathbf{b}$ are assembled from the local operators:

$$G = \sum_{k=1}^M A(P_k)^\top A(P_k), \quad \mathbf{b} = \sum_{k=1}^M A(P_k)^\top \mathbf{y}(P_k).$$

- Everything happens in the reflectivity space, of dimension M .
- G and its SVD: computed once per neighborhood, then reused. At inference, a new acquisition only needs $\mathbf{b}$ and 50 matrix-vector products.
- G depends on the position of the neighborhood, hence the grid of pre-trained networks.

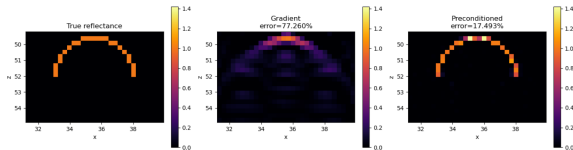
Appendix: Spectral Preconditioning

- Same fixed point, same least-squares solution: Q_ε only rescales the step per direction.

$$Q_\varepsilon = V \operatorname{diag}\left(\frac{1}{(\sigma_i + \varepsilon)^2}\right) V^\top, \quad \rho^{(k+1)} = \rho^{(k)} - \mu Q_\varepsilon (G\rho^{(k)} - \mathbf{b}).$$

- Contraction factor: $1 - \sigma_i^2/\sigma_{\max}^2$ without, $1 - \mu \sigma_i^2/(\sigma_i + \varepsilon)^2$ with.
At $\sigma_i/\sigma_{\max} = 10^{-2}$: 0.9999 versus ≈ 0.75 .
- ε sets where the acceleration stops, and keeps the correction bounded on the smallest directions.

Figure



Appendix: Inside a Layer, and the Learned Parameters

- Each layer applies three steps, with six learned scalars, *shared across layers*:

1. Preconditioned gradient step (μ_k, ε_k)

$$\mathbf{z}^{(k)} = \boldsymbol{\rho}^{(k)} - \mu_k \mathbf{V} \tilde{\mathbf{p}}^{(k)}, \quad \tilde{p}_i^{(k)} = \frac{\tilde{r}_i^{(k)}}{(\sigma_i + \varepsilon_k \sigma_{\max})^2}$$

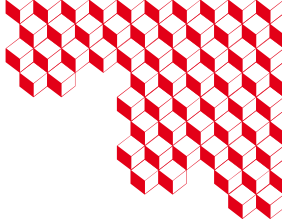
2. Adaptive thresholding (λ_k, T_k) : the smaller the coefficient, the stronger the threshold

$$\tilde{\rho}_i^{(k+1)} = \max \left(0, z_i^{(k)} - \lambda_k w_i^{(k)} \right), \quad w_i^{(k)} \propto \exp \left(- |z_i^{(k)}| / T_k \right)$$

3. Edge-preserving diffusion $(\gamma_k, T_k^{\text{sp}})$

$$\mathcal{D}_k(\mathbf{x})_i = x_i + \gamma_k \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(k)} (x_j - x_i), \quad \alpha_{ij}^{(k)} \propto \exp \left(- |x_i - x_j| / T_k^{\text{sp}} \right)$$

- Constraints imposed by *reparametrization*, so the optimization stays unconstrained: $\mu_k = \mu_{\max} \sigma(\mu_k^{\text{raw}})$, softplus for the others.
- Sharing across layers keeps a direct correspondence with the original algorithm; a per-layer variant would give $6K$ scalars.



Thank you!

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